



Report

Robot Navigation

Dinh-Son Vu

Table of Contents

1. Pre-requisites	3
2. Tasks	3
3. Explanations	4
4. Packages, nodes, and topics	4
5. Future works	4

1. Pre-requisites

- Ubuntu 24.04 Noble can run on your computer (VM, UTM, or dual boot).
- ROS2 Jazzy is installed.
- This is simulation only. The hardware is not required.

2. Tasks

Navigation corresponds to the movement of the robot in the environment, without the user controlling its trajectory. There are infinite paths for the robot to move to the final destination: in a straight line, swirling, going as far as possible from the wall, etc.

Tasks	Code
Install nav2	<code>sudo apt install ros-jazzy-navigation2 ros-jazzy-nav2-bringup -y</code>
Install twist-mux	<code>sudo apt install ros-jazzy-twist-mux</code>
Install twist_stamper	<code>sudo apt install ros-jazzy-twist-stamper</code>
Git clone the repo	<code>git clone https://github.com/ROS2-rmitbot/lesson5_ws.git</code>
Navigate to the workspace, build and source it	<pre>cd ~/lesson5_ws rm -rf build install log colcon build source install/setup.bash code .</pre>
Launch the bringup launch	<code>ros2 launch rmit_bringup rmitbot.launch.py</code>

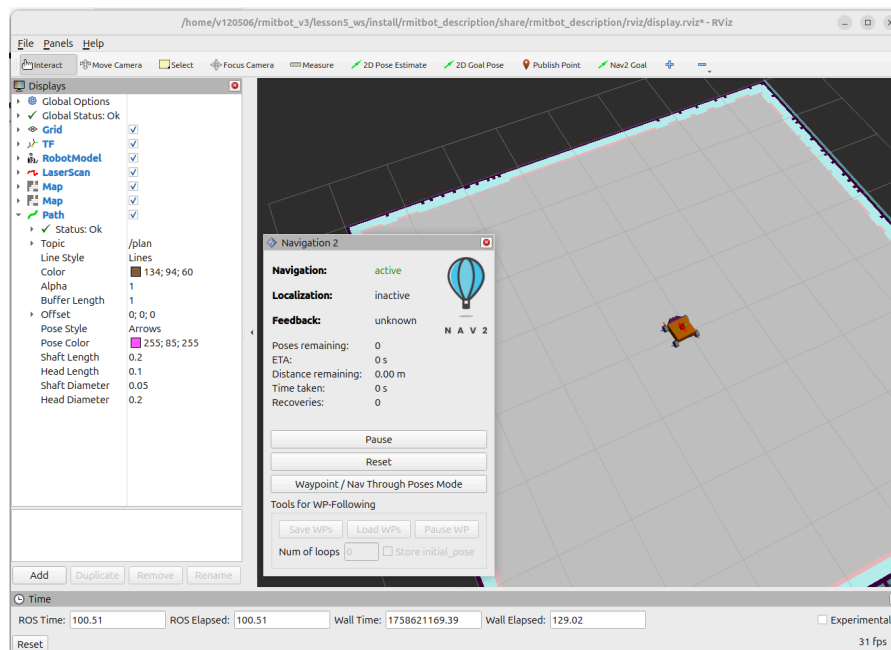


Figure 1: in Rviz, use the goal button to navigate the robot. Add obstacles in Gazebo and see the robot navigate.

3. Explanations

- A world has been added in the urdf.
- The package twistmux has been installed, to control the robot with the keyboard (high priority) and with nav2 (low priority).
- The package nav2 has been installed.
- In RViz, select a 2D Goal Pose: the robot should travel to the desired pose autonomously.
- Add some obstacles in Gazebo:
 - o The lidar should detect it.
 - o The robot should navigate and avoid the newly added obstacles.

Note that the robot can be controlled with the teleopkeyboard (as usual) and the navigation waypoint in rviz2.

The nav2 package includes several topics. You may check yourself the new topics and nodes launch with the nav2 package:

- ros2 node list: list of all the nodes launched
- ros2 topic list: list of all the topics

4. Packages, nodes, and topics

Packages	Nodes	Description
rmitbot_bringup		Launch the other packages and nodes
rmitbot_description	display.launch.py: rviz, rsp gazebo.launch.py: gazebo	Launch the urdf and static tf
rmitbot_controller	Controller manager: jsb spawner, controller spawner	Launch input/output of the robot
rmitbot_localization	ekf	Launch the sensor fusion between wheel encoder and imu
rmitbot_mapping	slam	Launch the lidar and slam
rmitbot_navigation	nav2	Launch the navigation

5. Future works

- The navigation yaml contains a lot of parameters to tune. It is up to the discretion of the student to fine tune the robot.